

Forward it. We'll tell you if the market actually said it.

Fake announcements, doctored filings and guaranteed-return schemes spread through chat faster than any regulator can chase them. TrustRail answers one question with a definite answer: **did a registered company really publish this?**

1.000

precision · recall · F1, measured

0

false positives across 81 eval
cases

28.9ms

median end-to-end verify latency

THE IDEA

Detection guesses. Provenance proves.

Most anti-fraud tools look at a message and estimate how suspicious it seems — a guess, and guesses can't be audited. TrustRail asks something answerable instead: does this match what a registered company actually published? It refuses to say "genuine" without either a valid signature or a match against that record.

HOW A VERDICT IS DECIDED

Three ordered checks. Every verdict traces to exactly one of them.

The engine short-circuits: it stops at the first stage that produces an answer. Nothing is ever called genuine without clearing stage 1 or stage 2 — stage 3 can only make things worse, never better.

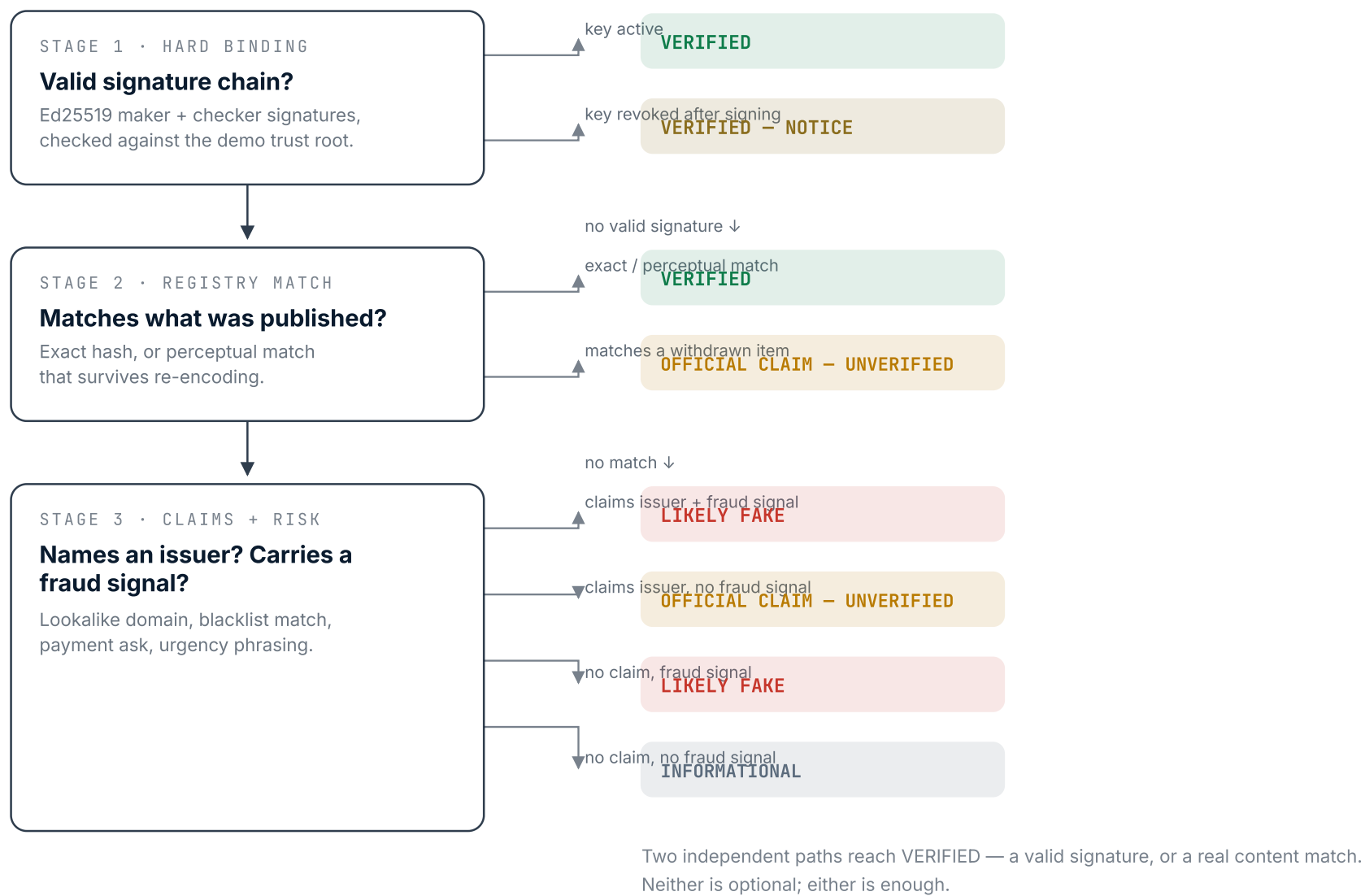


Figure 1. The exact decision order in `pipeline/verdict.py`: hard binding, then registry match, then claims + risk. Every one of the five verdicts traces back to a single labeled branch above — nothing is invented downstream.

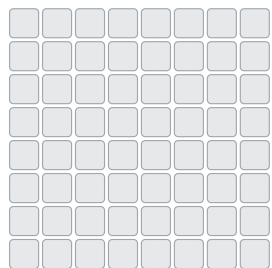
WHY STAGE 2 HAS TO EXIST

A forward rewrites every byte. It barely touches the picture.

Chat apps re-save every image and video they carry — that destroys any embedded signature, and a byte-for-byte hash check would miss the forward every time. TrustRail compares what the file *looks like* instead.

ORIGINAL, AS PUBLISHED

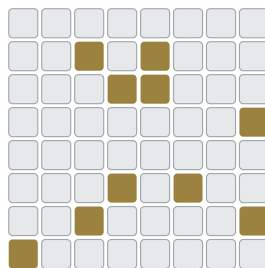
Byte hash (SHA-256)
a417e2b09c6f41d3...7cd1



Perceptual fingerprint (phash64) — each cell is one bit
distance 0 / 64

SAME IMAGE, AFTER A WHATSAPP FORWARD

Byte hash (SHA-256)
6b9033f8a1de5522...0ff4
completely different — useless for matching



10 of 64 bits differ — well inside the match threshold of 10
distance 10 / 64 → still a match

Figure 2. Illustrative pair matching the product’s own acceptance-suite example: a published notice image against a 2.5%-cropped, re-compressed copy. The fingerprint grid is not a diagram of the hash — a phash64 *is* an 8×8 grid of bits, so this is the hash itself.

THE FIVE VERDICTS

Strict about fraud. Never strict about doubt.

One real fraud signal is enough to escalate. Failing to find a match never is — that only ever means the answer is *cannot confirm*, not *fake*.

VERIFIED

Yes, this is genuine

Matches something the company really published.

VERIFIED_NOTICE

Genuine, but check first

Real when published — the signing key was later reported stolen.

OFFICIAL_CLAIM_UNVERIFIED

We cannot confirm this

Claims to be official; nothing published matches. Not an accusation.

LIKELY_FAKE

This looks like a scam

Carries fraud signals
TrustRail treats as serious.

INFORMATIONAL

Nothing official here

Makes no official claim at all. Most forwards land here.

PRODUCT SURFACES

Eight surfaces, one pipeline underneath.

Every page below calls the exact same verdict engine — there is no separate, weaker "demo mode."

01

Verify

Drop a file, text, or URL. A verdict in under a second, with a step-by-step trace and a full evidence panel.

02

Issuer console

Maker signs, checker co-signs, publish — the transparency log root visibly updates on that exact click.

03

Public record

Append-only transparency log. Your browser re-derives the Merkle inclusion proof itself, trusting no server.

04

Supervision

Regulator view: an India choropleth of where flags land, top-imitated entities, live campaign clustering.

05

Trust Circle + Auto-Guard

A guardian links to an elder’s phone. Every incoming message is scanned automatically — no forwarding, no clicking.

06

Channels

The identical pipeline through Telegram, WhatsApp and SMS reply formatting — not a web-only demo.

07

Registry

The public list of recognised issuers: official domains, SMS sender IDs, and live signing-key status.

08

One-time certificates

A shareable, single-use proof link per check — signature chain, log root, and inclusion proof.

Six moments, in order, each proving a different claim.

1

Publish live
A CEO video is signed and published on stage. The log root updates in real time.

2

Forward it, mangled
Same video, WhatsApp-re-encoded first.

VERIFIED

3

Doctor a filing
Real wording, one changed revenue figure.

LIKELY FAKE

4

Fake IPO SMS
Lookalike domain, tied to a tracked campaign.

LIKELY FAKE

5

Plain news text
No claim at all — a calm, honest non-alarm.

INFORMATIONAL

6

Revoke & re-check
Same video as step 2, after the signing key is revoked.

VERIFIED, NOTICE

MEASURED, NOT CLAIMED

Evaluated against 81 cases — 31 positive, 50 negative.

Every transform a real WhatsApp forward actually applies: JPEG re-compression, resizing, metadata stripping, screenshotting, video re-encoding, zero-width character stripping, emoji injection.

1.000

Precision across all 81 cases

1.000

Recall — held on every single transform

1.000

F1 — 0 false positives out of 50 negatives

TRANSFORM APPLIED	RECALL	CASES
Image · JPEG q50 / q70 / q85	1.00	3/3 each
Image · resize 0.5× / 0.75×, metadata strip, screenshot sim	1.00	3/3 each
Video · CRF26 / CRF30 re-encode	1.00	1/1 each
Text · whitespace/emoji injection, zero-width strip	1.00	4/4 each
End-to-end latency	p50 28.9ms	p95 48.6ms

BUILT WITH

Tech stack

Backend

Python / FastAPI

PostgreSQL

Redis

SQLAlchemy + Alembic

PyNaCl (Ed25519)

Pillow / imagehash

ffmpeg

rapidfuzz

Frontend

Next.js 14

TypeScript

Tailwind CSS

Recharts

NON-NEGOTIABLE

Guardrails

- ✓

A test enforces that no code path can emit **Verified** without a valid signature or a registry match — a release blocker if violated.
- ✓

No invented numbers anywhere in the UI — every stat rendered comes from the database.
- ✓

No ML confidence score, ever. When provenance is absent, the product says so honestly instead.

react-simple-maps

Trust primitives

Ed25519 signature envelope

RFC 6962 Merkle log

Client-side inclusion proof

- ✔ Banned in UI copy: "blockchain," "crypto," "ledger," "multi-sig." Plain terms only.
- ✔ Fictional entities only — no real brands, tickers, or persons anywhere in fixtures or copy.

TrustRail: hackathon prototype, not affiliated with SEBI.

All entities, filings and campaigns shown are fictional. The cryptography is real.